Solution Requirements

Phase 2 | Functional scope and quality attributes

Functional requirements

ID	Requirement	Implemented route or screen
FR-01	Register, sign in, and load current profile.	/auth/register, /auth/login, /auth/me; Account page.
FR-02	Generate an itinerary from trip preferences.	POST /trips/plan; CreateTrip page.
FR-03	Save, list, retrieve, edit, and delete trips.	POST /trips/create; GET/PATCH/DELETE /trips routes.
FR-04	Regenerate a selected itinerary day using feedback.	POST /trips/regenerate-day; TripView page.
FR-05	Copy/share itinerary content or download Markdown.	TripView browser actions.
FR-06	Support a local structured generation fallback.	AI orchestrator and destination data services.

Quality requirements

- Security: signed expiring bearer tokens, password hashing, protected routes, ownership checks, and server-only service-role key.
- Reliability: validated response models and local generation fallback.
- Maintainability: separate UI pages, API client, routers, schemas, and services.
- Deployability: environment configuration and Docker/Render/Vercel setup.
- Usability: loading and error states for account, planning, and saved-trip flows.

Constraints and exclusions

Date values are strings in request schemas; strict date ordering should be verified separately. In-memory persistence is temporary. Booking, real-time availability, maps, and anonymous public sharing are outside current source behavior.

Acceptance criteria

- A valid request returns trip fields, day objects, and activities.
- A signed-in user can save and retrieve a trip with its itinerary.
- Trip operations respect the authenticated owner's identity.
- When remote generation fails, the local itinerary path can return structured output.
- A regenerated day can be persisted while retaining the other days.

Prepared from the current project source and configuration. Product assumptions and proposed work are labeled; no user research, test result, or deployment result is implied unless explicitly stated.